Matija Akrap

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GitHub | LinkedIn

Education

KU Leuven — Leuven, Belgium
Exchange student, MSc year, Faculty of Engineering Science — *Sep 2026 – Sep 2027*
- Coursework: Machine Learning, NLP, Computer Vision, Reinforcement Learning, Bio-informatics etc.

University of Zagreb — Zagreb, Croatia
MSc in Data Science, Faculty of Electrical Engineering and Computing (FER) — *Oct 2026 – Jul 2028 (exp.)*
BSc in Computing — *Oct 2023 – Jul 2026*
- Thesis: *Characterizing Research Paper Promotion on Social Media* (supervisor: Prof. Jan Šnajder).

Research Interests

Neurosymbolic AI, reasoning and evaluation of LLMs, reinforcement learning, mechanistic interpretability.

Publications

M. Akrap, A. Bilić, R. Šimpraga, F. Račić, L. Čuturilo. FER at SemEval-2026 Task 6: Analysis of Different Approaches to Unmasking Political Question Evasions. *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval-2026)*, co-located with ACL 2026, pages 30–36. [Paper] [GitHub]

Research

Detecting Evasive Answers in Political Interviews — Oct 2025 – Jun 2026
SemEval-2026 Task 6, research project in the Project R course, FER — *Zagreb, Croatia*
- Compared flat, hierarchical and multitask RoBERTa models, with the hierarchical model performing best.
- Found that multitask models broke the label hierarchy, and enforcing it at inference raised F1.

Projects

Personal Language Model | *PyTorch, QLoRA, mlx-lm* [GitHub]
- Built a training set from my own WhatsApp and Instagram messages.
- Pre-trained a GPT-style model from scratch, then QLoRA fine-tuned Ministral-8B to write in my style.

Dobby: Movie & TV Recommendation Platform | *PyTorch, pgvector* [GitHub]
- Led a 7-person team building a social platform for rating, tracking and recommending films and shows.
- Built a hybrid recommender combining PyTorch matrix factorization with genre embeddings on 23.6M ratings.
- Solved user cold start by mapping genre picks into the embedding space, with pgvector and fuzzy-logic ranking.

Last Translation Benchmark | *contributor* [Paper]
- Wrote EN–HR cases using dialects, false friends and numerals that break 8+ of 10 top MT systems.

Industry and Teaching Experience

Frontend Developer — Jun 2025 – Jan 2026
Erste Group Card Processor — *Zagreb, Croatia*
- Shipped features across three internal banking apps in Angular and TypeScript, integrated with Spring Boot.
- Designed features end to end, from UI to backend integration, and fixed production bugs reported by staff.

Student Demonstrator (Teaching Assistant) — Oct 2023 – Feb 2025
FER: Digital Logic and Fundamentals of Electrical Engineering — *Zagreb, Croatia*
- Ran weekly lab sessions and graded practical assignments.

Skills

Programming: Python, C/C++, Java, R, SQL, TypeScript, JavaScript
ML & Data: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, fine-tuning
Engineering: Git, Linux, Bash, Angular, React, Spring Boot
Spoken languages: Croatian (native), English (C1)